

Unpaired and Unsupervised: A Dual-Model Framework for SAR Image Colorization and Downstream Building Verification

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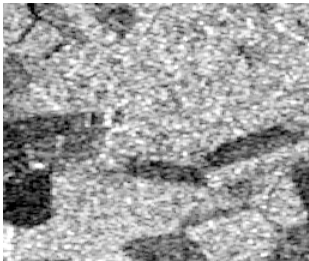
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Introduction

SAR Images (Synthetic Aperture Radar):

- Radar-based images captured using microwave signals.
- Works in all weather and lighting conditions.
- Grayscale and speckled; difficult to interpret visually.



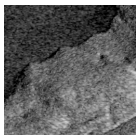
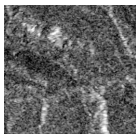
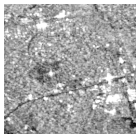
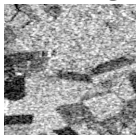
Optical Images:

- Satellite images captured in visible light, showing natural colors.
- Easy to understand but affected by clouds and daylight.



Introduction: Unsupervised and Unpaired Learning

SAR Images



Optical Images



SAR Interpretability Challenges

- SAR images are difficult to interpret visually.
- Lack of natural colors and contain complex speckle patterns.

Limitations of Optical Images

- Optical images often unavailable due to clouds or night.
- This limits their use in continuous monitoring.

Bridging Sensor Modalities

- Colorizing SAR improves human interpretation and downstream tasks.
- Unpaired methods allow scaling without scarce paired datasets.

Objective

- Design an unsupervised and unpaired learning framework for SAR→Optical translation.
- Evaluate impact on downstream building verification (UNet segmentation).
- Compare raw SAR, colorized SAR, and real optical for downstream metrics.
- Analyze failure modes and propose mitigations for speckle, color shift, and resource constraints.

1. *“An unpaired SAR-to-optical image translation method based on Schrödinger bridge network and multi-scale feature fusion”* (Wang et al., 2024) [1]

- Introduces a Schrödinger-bridge based domain-transfer framework combined with a multiscale axial residual module (MARM) for unpaired SAR → optical translation.
- Demonstrates major gains vs classical CycleGAN
- Highlights challenges of SAR images (speckle noise, geometric distortion) and the lack of paired training data, motivating the unpaired approach.

2. “SAR-to-RGB Translation with Latent Diffusion for Earth Observation” (Aydin et al., 2025) [2]

- Proposes a latent-space diffusion model to translate SAR (S1) $\rightarrow$ RGB (S2) imagery for Earth observation tasks.
- Evaluates the generated images not only by visual fidelity (SSIM, FID) but also via downstream tasks: land-cover classification and cloud removal.
- Finds that although perceptual quality may lag real RGB, synthetic outputs still add value in downstream tasks; emphasizes that typical metrics may not fully capture practical utility.

Methodology — CycleGAN (expanded)

- **Unpaired Image Translation:**

Learns $\text{SAR} \leftrightarrow \text{RGB}$ mapping without paired datasets.

- **Two Generators (G & F):**

- $G: \text{SAR} \rightarrow \text{RGB}$

- $F: \text{RGB} \rightarrow \text{SAR}$

Translate images across domains.

- **Two Discriminators (D_X & D_Y):**

PatchGAN-based networks that classify local patches as real or fake.

- **Adversarial Loss:**

Encourages generators to produce outputs that fool the discriminators.

- **Cycle-Consistency Loss:**

Ensures $\text{SAR} \rightarrow \text{RGB} \rightarrow \text{SAR}$ reconstruction stays close to the original.

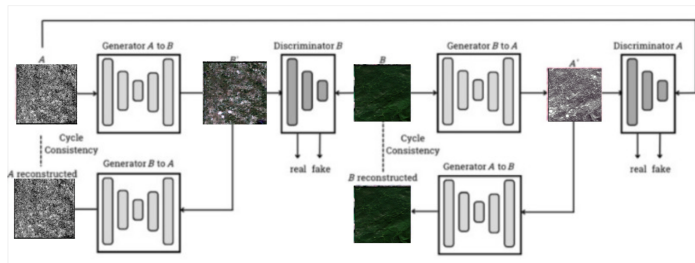
- **Identity Loss:**

Helps stabilize training and maintain color/brightness.

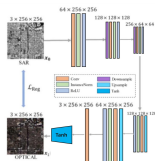
- **Final Output:**

RGB-like colored SAR images with preserved geometry and realistic textures.

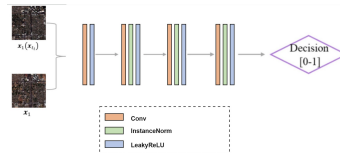
Model Architectures



CycleGAN



Generator



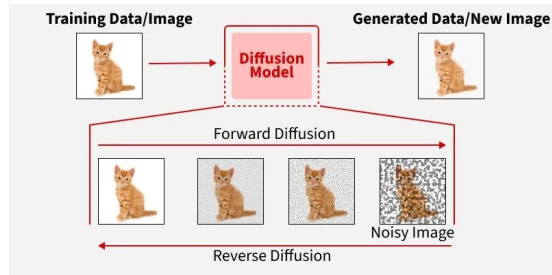
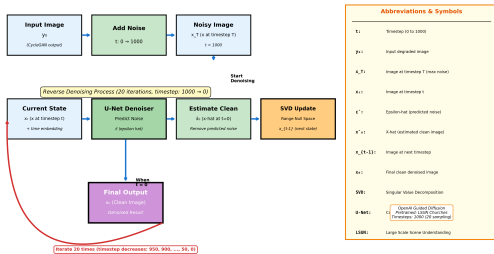
Discriminator

Methodology — OpenAI's pretrained guided Diffusion model

- **Forward Diffusion Process:** CycleGAN output undergoes progressive Gaussian noise addition over 1000 timesteps to create a maximally noisy image.
- **OpenAI's Guided Diffusion U-Net:** Pretrained denoiser on LSUN (Large Scale Scene Understanding) Churches dataset predicts noise from current noisy state conditioned on timestep embedding.
- **Clean Image Estimation:** Predicted noise is subtracted from current state to estimate the clean image at each denoising iteration.
- **SVD (Singular Value Decomposition) Range-Null Space:** Combines input-guided range space with diffusion-hallucinated null space to compute the next cleaner state.
- **DDIM (Denoising Diffusion Implicit Models) Sampling:** Iterative reverse process with 20 accelerated steps and decreasing timesteps from 950 to 0 produces the final denoised output.

Diffusion Model

DDRM Diffusion Model Architecture

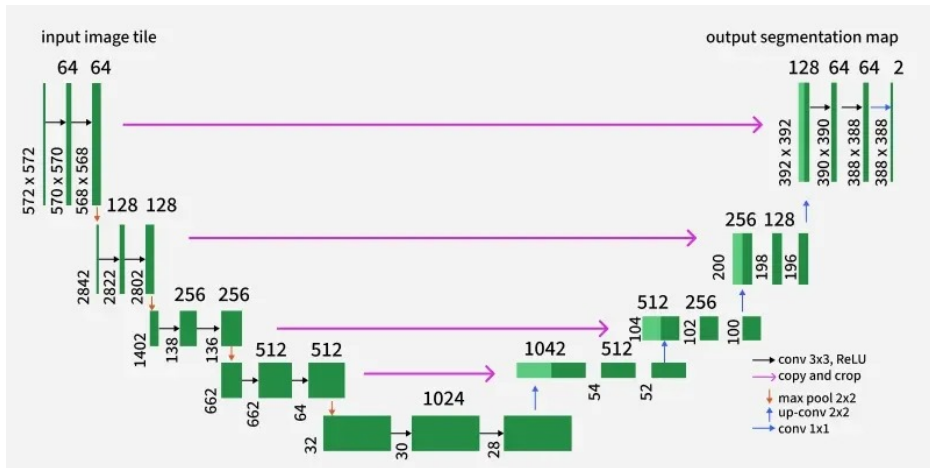


DDRM (Denoising Diffusion Restoration Model)[3]

Methodology — UNet for Downstream Building Verification

- No manual masks or ground-truth used.
- Pixel features extracted using pretrained CNN (ResNet50).
- K-Means clustering used to group pixels.
- Building cluster chosen automatically and noise removed.
- Pseudo-masks used to train a U-Net.
- U-Net refines and outputs clean building segmentation maps.

UNet Architecture (diagram)



UNet Model [4]

Dataset Used: Sentinel-1 and Sentinel-2 Image Pairs [5]

Sentinel-1 (SAR Images)

- Agri – 4000 images
- Barrenland – 4000 images
- Urban – 4000 images
- Grassland – 4000 images

Pixel Values: 256x256, Single Band

Total: 16000 images

Avg. size: 60KB per image

Sentinel-2 (Optical Images)

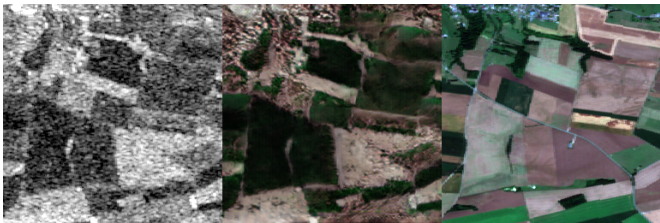
- Agri – 4000 images
- Barrenland – 4000 images
- Urban – 4000 images
- Grassland – 4000 images

Pixel Values: 256x256, 3 Bands

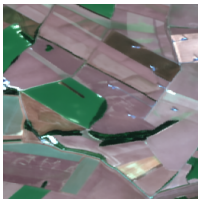
Total: 16000 images

Avg. size: 125KB per image

Results — Model Outputs



CycleGAN

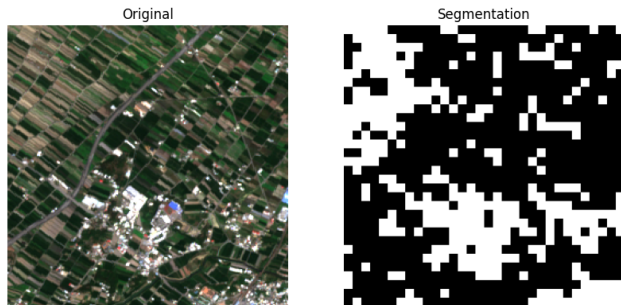


DDRM



DDRM

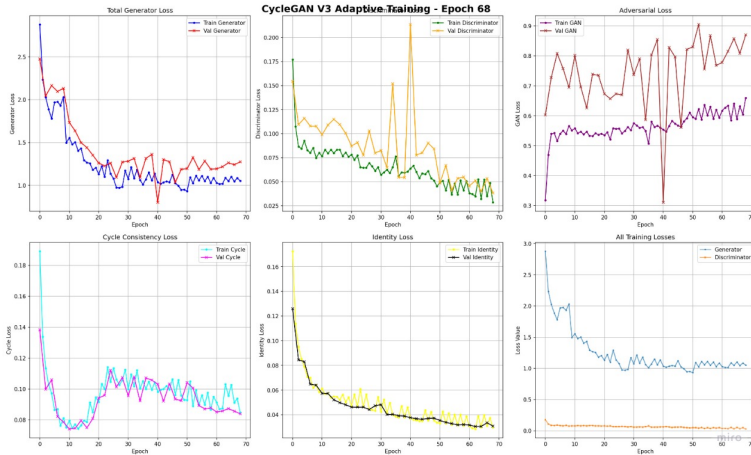
Results



UNet

Model	PSNR (dB)	SSIM	Test Samples
CycleGAN (7 Residual Blocks)	10.33 ± 1.28	0.104 ± 0.029	200
CycleGAN (5 Residual Blocks)	9.48 ± 1.66	0.027 ± 0.029	200

Results



Loss vs. Epoch Graph

Setbacks & Challenges

- **Initial CycleGAN Attempt with 9 Residual Blocks:**

Google Colab had strict resource limits, and the training speed was extremely slow (only 1 epoch per day), making experimentation difficult.

- **Hardware Limitations Across Multiple Systems:**

The university HPC was fully occupied. Training on an RTX A400 (4GB) showed that 20 epochs of CycleGAN would take around 11–12 days. Even after running for 2 days, progress remained slow. We later switched to an RTX A1000 (8GB) for further training.

- **Overfitting Issues:**

Training CycleGAN with 9 residual blocks caused strong overfitting. Each epoch required nearly 2 hours, and after around 10 epochs the model clearly overfit. We updated learning rates for both generator and discriminator, added validation splits, and tracked PSNR and SSIM, but the results were still unsatisfactory.

Setbacks & Challenges

- **Power Interruptions:**

Since the training was executed on university systems, several power cuts interrupted the process. This forced us to restart training multiple times, increasing the total time required.

- **Architectural and Hyperparameter Adjustments:**

We reduced generator complexity from 9 to 7 residual blocks, and later to 5. We tuned multiple hyperparameters, but improvements remained limited. Training with 5 residual blocks is currently ongoing.

- **Exploring Alternative Approaches:**

Due to limited performance from CycleGAN, we explored OpenAI's pretrained diffusion model (LSUN Churches). We also built a building segmentation pipeline using U-Net trained on pseudo-masks.

References I

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-  K. Aydin, J. Hanna, and D. Borth, "Sar-to-rgb translation with latent diffusion for earth observation," *arXiv preprint arXiv:2504.11154*, 2025.
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Thank You !!!!